Topics Covered:

- Recap: Computing Jacobians
- Individual Body Terms
- Manipulator Types
- Practice Problems

Additional Reading:

- LP 5.3

Recap: Computing Jacobians

A Jacobian maps joint velocities $\dot{\theta}$ to end-effector velocity. Which Jacobian we compute depends on how we describe the end-effector.

Coordinate Jacobian

If the end-effector is described by coordinates $x = f(\theta)$ (for example, only its position), then we differentiate the forward kinematics directly:

$$\dot{x} = J(\theta)\dot{\theta}, \quad \text{with} \quad J(\theta) = \frac{\partial f(\theta)}{\partial \theta} = \begin{bmatrix} \frac{\partial f_1(\theta)}{\partial \theta_1} & \cdots & \frac{\partial f_1(\theta)}{\partial \theta_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(\theta)}{\partial \theta_1} & \cdots & \frac{\partial f_m(\theta)}{\partial \theta_n} \end{bmatrix}$$

Manipulator Jacobian

If the end-effector is described by its full pose $g_e(\theta)$, then its velocity is a twist. The spatial and body manipulator Jacobians map joint velocities to the spatial and body twists:

$$\xi_s = J_s(\theta)\dot{\theta}, \qquad \xi_b = J_b(\theta)\dot{\theta}, \qquad J_s(\theta) = \mathrm{Ad}_{g_e(\theta)} J_b(\theta)$$

Recall that this adjoint operation can be applied either column-wise or using the matrix form of the adjoint:

$$\mathrm{Ad}_g \xi = \left(g\hat{\xi}g^{-1}\right)^{\vee}, \qquad \mathrm{Ad}_g = \begin{bmatrix} R & [d]_\times R \\ 0 & R \end{bmatrix}$$

Ways to Compute the Manipulator Jacobian

1. **From the definition.** Differentiate the full forward kinematics:

$$J_s(\theta) = \begin{bmatrix} \left(\frac{\partial g_e}{\partial \theta_1} g_e^{-1}\right)^\vee & \cdots & \left(\frac{\partial g_e}{\partial \theta_n} g_e^{-1}\right)^\vee \end{bmatrix}$$
$$J_b(\theta) = \begin{bmatrix} \left(g_e^{-1} \frac{\partial g_e}{\partial \theta_1}\right)^\vee & \cdots & \left(g_e^{-1} \frac{\partial g_e}{\partial \theta_n}\right)^\vee \end{bmatrix}$$

2. **By inspection.** Write each joint twist at the *current* configuration:

$$J_s(\theta) = \begin{bmatrix} \xi_1 & \xi_2' & \cdots & \xi_n' \end{bmatrix}, \quad \text{with} \quad \xi_i' = \begin{cases} \begin{bmatrix} -\hat{\omega}_i' \times q_i' \\ \hat{\omega}_i' \end{bmatrix} & \text{if revolute} \\ \begin{bmatrix} v_i' \\ 0 \end{bmatrix} & \text{if prismatic} \end{cases}$$

3. **Using adjoints of the zero-configuration twists.** Given the product of exponentials $g_e = e^{\hat{\xi}_1 \theta_1} \cdots e^{\hat{\xi}_n \theta_n} g_0$:

$$J_s(\theta) = \begin{bmatrix} \xi_1 & \mathrm{Ad}_{e^{\hat{\xi}_1 \theta_1}} \xi_2 & \cdots & \mathrm{Ad}_{e^{\hat{\xi}_1 \theta_1} \cdots e^{\hat{\xi}_{n-1} \theta_{n-1}}} \xi_n \end{bmatrix}$$
$$J_b(\theta) = \begin{bmatrix} \mathrm{Ad}^{-1}_{e^{\hat{\xi}_1 \theta_1} \cdots e^{\hat{\xi}_n \theta_n} g_0} \xi_1 & \cdots & \mathrm{Ad}^{-1}_{e^{\hat{\xi}_n \theta_n} g_0} \xi_n \end{bmatrix}$$

4. **Using individual body terms.** Differentiate one link at a time, then transform each term with an adjoint. Given the product of Lie groups $g_e = g_1 \cdots g_n g_{n+1}$:

$$J_s(\theta) = \begin{bmatrix} J_{s,1} & \mathrm{Ad}_{g_1} J_{s,2} & \cdots & \mathrm{Ad}_{g_1 \cdots g_{n-1}} J_{s,n} \end{bmatrix}$$
$$J_b(\theta) = \begin{bmatrix} \mathrm{Ad}^{-1}_{g_2 \cdots g_{n+1}} J_{b,1} & \cdots & \mathrm{Ad}^{-1}_{g_{n+1}} J_{b,n} \end{bmatrix}$$

with $J_{s,i} = \left(\frac{\partial g_i}{\partial \theta_i} g_i^{-1}\right)^\vee$ and $J_{b,i} = \left(g_i^{-1} \frac{\partial g_i}{\partial \theta_i}\right)^\vee$. This is covered in the next section.

Recap Example

Consider a planar arm with two revolute joints and link lengths l_1 and l_2:

$$g_e(\theta) = g_1(\theta_1) g_2(\theta_2) g_3, \quad \text{with} \quad g_1 = \begin{bmatrix} R(\theta_1) & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}, \quad g_2 = \begin{bmatrix} R(\theta_2) & \begin{bmatrix} l_1 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}, \quad g_3 = \begin{bmatrix} I & \begin{bmatrix} l_2 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

Since this arm is planar, its twists have the form $\xi = [v_x, v_y, \omega]^\top$.

Both joint axes point out of the page. By inspection, the second joint is currently located

at $q_2' = [l_1\cos(\theta_1), l_1\sin(\theta_1)]^\top$, which gives:

$$J_s(\theta) = \begin{bmatrix}\xi_1 & \xi_2'\end{bmatrix} = \begin{bmatrix}0 & l_1\sin(\theta_1)\\ 0 & -l_1\cos(\theta_1)\\ 1 & 1\end{bmatrix}$$

Using adjoints instead, the zero-configuration twists are $\xi_1 = [0,0,1]^\top$ and $\xi_2 = [0,-l_1,1]^\top$. Since ξ_1 is a pure rotation about the origin, its exponential is:

$$e^{\hat{\xi}_1\theta_1} = \begin{bmatrix}\cos(\theta_1) & -\sin(\theta_1) & 0\\ \sin(\theta_1) & \cos(\theta_1) & 0\\ 0 & 0 & 1\end{bmatrix}$$

The second column, ξ_2', is then equivalently computed as:

$$\begin{aligned}\mathrm{Ad}_{e^{\hat{\xi}_1\theta_1}}\xi_2 &= \left(e^{\hat{\xi}_1\theta_1}\hat{\xi}_2 e^{-\hat{\xi}_1\theta_1}\right)^\vee\\ &= \left(\begin{bmatrix}\cos(\theta_1) & -\sin(\theta_1) & 0\\ \sin(\theta_1) & \cos(\theta_1) & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & -1 & 0\\ 1 & 0 & -l_1\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}\cos(\theta_1) & \sin(\theta_1) & 0\\ -\sin(\theta_1) & \cos(\theta_1) & 0\\ 0 & 0 & 1\end{bmatrix}\right)^\vee\\ &= \begin{bmatrix}0 & -1 & l_1\sin(\theta_1)\\ 1 & 0 & -l_1\cos(\theta_1)\\ 0 & 0 & 0\end{bmatrix}^\vee = \begin{bmatrix}l_1\sin(\theta_1)\\ -l_1\cos(\theta_1)\\ 1\end{bmatrix}\end{aligned}$$

We can also use the 6×6 matrix form of the adjoint from Lecture 15. This form acts on six-dimensional twists, so we first write the same joint twist in 3D as $\xi_2 = [0,-l_1,0,0,0,1]^\top$. In 3D, $e^{\hat{\xi}_1\theta_1}$ has rotation $R = R_z(\theta_1)$ and translation $d = 0$, which means that $[d]_\times R = 0$. Plugging these in:

$$\begin{aligned}\mathrm{Ad}_{e^{\hat{\xi}_1\theta_1}}\xi_2 &= \begin{bmatrix}R & [d]_\times R\\ 0 & R\end{bmatrix}\xi_2\\ &= \begin{bmatrix}\cos(\theta_1) & -\sin(\theta_1) & 0 & 0 & 0 & 0\\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & \cos(\theta_1) & -\sin(\theta_1) & 0\\ 0 & 0 & 0 & \sin(\theta_1) & \cos(\theta_1) & 0\\ 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0\\ -l_1\\ 0\\ 0\\ 0\\ 1\end{bmatrix} = \begin{bmatrix}l_1\sin(\theta_1)\\ -l_1\cos(\theta_1)\\ 0\\ 0\\ 0\\ 1\end{bmatrix}\end{aligned}$$

Keeping only the planar components $[v_x, v_y, \omega]^\top$ gives the same second column.

As expected, both methods give the same spatial manipulator Jacobian.

Individual Body Terms

We can derive one final method of obtaining the manipulator Jacobians using the product of Lie groups. This will allow us to write the equation in terms of the following **body terms**:

$$J_{b,i}(\theta_i) = \left(g_i^{-1}\frac{\partial g_i}{\partial \theta_i}\right)^\vee,$$
$$J_{s,i}(\theta_i) = \left(\frac{\partial g_i}{\partial \theta_i}g_i^{-1}\right)^\vee = \mathrm{Ad}_{g_i} J_{b,i}(\theta_i).$$

The two are related by an adjoint since $g_i\left(g_i^{-1}\frac{\partial g_i}{\partial \theta_i}\right)g_i^{-1} = \frac{\partial g_i}{\partial \theta_i}g_i^{-1}$.

Derivation using Product of Lie Groups

Let's start with the two-joint arm from our recap example, where $g_e = g_1(\theta_1)g_2(\theta_2)g_3$. Since each transformation is only a function of its own joint variable:

$$\frac{\partial g_e}{\partial \theta_1} = \frac{\partial g_1}{\partial \theta_1}g_2 g_3, \qquad \frac{\partial g_e}{\partial \theta_2} = g_1\frac{\partial g_2}{\partial \theta_2}g_3$$

Plugging these into the definition of the body manipulator Jacobian (before unhatting each column):

$$\begin{aligned} J_b(\theta) = g_e^{-1}\frac{\partial g_e}{\partial \theta} &= g_3^{-1}g_2^{-1}g_1^{-1}\begin{bmatrix}\frac{\partial g_1}{\partial \theta_1}g_2 g_3 & g_1\frac{\partial g_2}{\partial \theta_2}g_3\end{bmatrix} \\ &= \begin{bmatrix} g_3^{-1}g_2^{-1}\underbrace{g_1^{-1}\frac{\partial g_1}{\partial \theta_1}}_{\text{body term}}g_2 g_3 & g_3^{-1}\underbrace{g_2^{-1}\cancel{g_1^{-1}}\cancel{g_1}\frac{\partial g_2}{\partial \theta_2}}_{\text{body term}}g_3 \end{bmatrix} \\ &= \begin{bmatrix}(g_2g_3)^{-1}(J_{b,1})^\wedge(g_2g_3) & g_3^{-1}(J_{b,2})^\wedge g_3\end{bmatrix} \\ &= \begin{bmatrix}\mathrm{Ad}_{g_2g_3}^{-1}J_{b,1} & \mathrm{Ad}_{g_3}^{-1}J_{b,2}\end{bmatrix} \end{aligned}$$

The same cancellation happens for an arm with any number of joints and yields the general formula:

$$J_b(\theta) = \begin{bmatrix}\mathrm{Ad}_{g_2\cdots g_{n+1}}^{-1}J_{b,1} & \cdots & \mathrm{Ad}_{g_{n+1}}^{-1}J_{b,n}\end{bmatrix}$$

To derive the spatial manipulator Jacobian, we apply $J_s = \mathrm{Ad}_{g_e}J_b$. Using $\mathrm{Ad}_g\mathrm{Ad}_h^{-1} = \mathrm{Ad}_{gh^{-1}}$, the kth column is:

$$\begin{aligned} \mathrm{Ad}_{g_e}\mathrm{Ad}^{-1}_{\prod_{j=k+1}^{n+1}g_j}J_{b,k} &= \mathrm{Ad}_{(g_1\cdots g_{n+1})(g_{k+1}\cdots g_{n+1})^{-1}}J_{b,k} \\ &= \mathrm{Ad}_{g_1\cdots g_k}J_{b,k} \\ &= \mathrm{Ad}_{g_1\cdots g_{k-1}}\mathrm{Ad}_{g_k}J_{b,k} \\ &= \mathrm{Ad}_{g_1\cdots g_{k-1}}J_{s,k} \end{aligned}$$

which yields the general formula for J_s:

$$J_s(\theta) = \begin{bmatrix} J_{s,1} & \mathrm{Ad}_{g_1} J_{s,2} & \cdots & \mathrm{Ad}_{g_1 \cdots g_{n-1}} J_{s,n} \end{bmatrix}$$

Example

Let's compute the spatial manipulator Jacobian for our two-joint arm using the body terms:

$$\begin{aligned}
(J_{s,1})^\wedge &= \frac{\partial g_1}{\partial \theta_1} g_1^{-1} = \begin{bmatrix} -\sin(\theta_1) & -\cos(\theta_1) & 0 \\ \cos(\theta_1) & -\sin(\theta_1) & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta_1) & \sin(\theta_1) & 0 \\ -\sin(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
(J_{s,2})^\wedge &= \frac{\partial g_2}{\partial \theta_2} g_2^{-1} = \begin{bmatrix} -\sin(\theta_2) & -\cos(\theta_2) & 0 \\ \cos(\theta_2) & -\sin(\theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta_2) & \sin(\theta_2) & -l_1 \cos(\theta_2) \\ -\sin(\theta_2) & \cos(\theta_2) & l_1 \sin(\theta_2) \\ 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -l_1 \\ 0 & 0 & 0 \end{bmatrix}
\end{aligned}$$

Thus, $J_{s,1} = [0, 0, 1]^\top$ and $J_{s,2} = [0, -l_1, 1]^\top$. Next, taking the adjoint of the second term:

$$\mathrm{Ad}_{g_1} J_{s,2} = \left(g_1 \left(J_{s,2} \right)^\wedge g_1^{-1} \right)^\vee = \begin{bmatrix} 0 & -1 & l_1 \sin(\theta_1) \\ 1 & 0 & -l_1 \cos(\theta_1) \\ 0 & 0 & 0 \end{bmatrix}^\vee = \begin{bmatrix} l_1 \sin(\theta_1) \\ -l_1 \cos(\theta_1) \\ 1 \end{bmatrix}$$

Putting these together gives the spatial manipulator Jacobian:

$$J_s(\theta) = \begin{bmatrix} J_{s,1} & \mathrm{Ad}_{g_1} J_{s,2} \end{bmatrix} = \begin{bmatrix} 0 & l_1 \sin(\theta_1) \\ 0 & -l_1 \cos(\theta_1) \\ 1 & 1 \end{bmatrix}$$

which matches the result from our recap example.

Manipulator Types (based on Workspace)

Note: this description of manipulator types is more functional, less geometric than before.

Let n = dimension of task space, m = dimension of joint space. In this case, $J \in \mathbb{R}^{n \times m}$.

The three (functional) classes to describe a given manipulator are:

1. kinematically insufficient ($n > m$)
   - can get a "better" manipulator so that $m = n$.
   - can reduce dimension of task space by redefining what the task is.
2. kinematically redundant ($n < m$)
   - have more degrees of freedom than needed (Inverse kinematics have infinite solutions).
   - it's tricky, but this case can be dealt with.
3. kinematically sufficient ($n = m$)
   - need to ensure operation in dextrous workspace for maximal mobility/control.
   - equivalently, need to be away from singularities (will discuss this now)

Definition: Singularities

A **singular configuration** or **singularity** is a joint configuration of an open-chain manipulator in which the end-effector instantaneously loses a degree of freedom of its motion capability versus the number of degrees of freedom that normally prevail (i.e., the end-effector loses its ability to move in one or more directions).

Note here that open-chain means that joint-link-joint... combinations do *not* form a loop.

Practical implications of singularities:

- manipulator loses effectiveness
- high joint velocities may be needed near a singular configuration in order to track a specified trajectory
- the manipulator can have high mechanical advantage in a singular configuration. (bench pressing in a gym as your arm is fully extended)

Finding singular configurations:

To find singularities, we can examine the manipulator Jacobian:

1. Numerical Approach

$$\xi = J(\theta)\dot{\theta}$$

   Test the rank of J (the number of linearly independent columns)

2. For kinematically sufficient arms,

$$\det(J(\theta)) = 0$$

   at a singular configuration.

3. For redundant arms ($n < m$), $J(\theta)J^\top(\theta)$ has the same rank as $J(\theta)$ but is square. That means a perfectly valid test is:

$$\det(J(\theta)J^\top(\theta)) = 0$$

   for singular configurations.

4. For kinematically insufficient arms ($n > m$), $J^\top(\theta)J(\theta)$ has the same rank as $J(\theta)$ but is square. Thus, a test for singular configurations is $\det(J^\top J) = 0$.

Note, that for condition 4, the old notes says that there is an "issue" with this point, because one should not really even have this case in practice. If an arm is insufficient, it is best to get a new arm or modify the task to be sufficient. Also, the product $J^\top J$ has nonsense units if dimensional analysis is performed.

To find the singular configurations means applying one of the given tests for all θ in the joint space and keeping track of the singular ones.

There is connection between loss of rank in the Jacobian, and loss of control.

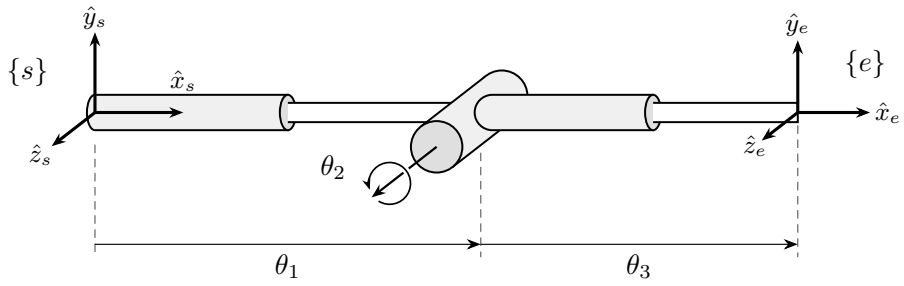
There are usually two occasions when a singularity occurs:

1. Workspace - bounded singularities
   - when manipulator is fully extended or folded back on itself in such a way that the end-effector is at, or near, the workspace boundary
2. Workspace - interior singularities
   - occur away from workspace boundary, generally caused by a lining up of two or more axes.

Practice Problems

Problem 1: Planar PRP Manipulator

Consider the planar manipulator below, shown in its zero configuration.



(a) Use the product of Lie groups to find the forward kinematics $g_e(\theta)$.

(b) Compute the body manipulator Jacobian $J_b(\theta)$.

(c) Evaluate $J_b(\theta)$ at $\theta = (1, \pi/2, 1)^\top$ and at $\theta = (1, 0, 1)^\top$. Is either configuration singular?

(d) For $\theta = (1, \theta_2, 1)^\top$, find all values of θ_2 that give a singular configuration.

Solution

(a) This manipulator has the forward kinematics (using product of Lie groups):

$$g_e(\theta) = g_1(\theta_1)g_2(\theta_2)g_3(\theta_3)$$

with

$$g_1 = \begin{bmatrix} I & \begin{bmatrix}\theta_1 \\ 0\end{bmatrix} \\ 0 & 1 \end{bmatrix}, \quad g_2 = \begin{bmatrix} R(\theta_2) & \begin{bmatrix}0 \\ 0\end{bmatrix} \\ 0 & 1 \end{bmatrix}, \quad g_3 = \begin{bmatrix} I & \begin{bmatrix}\theta_3 \\ 0\end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

Once multiplied, this gives us:

$$g_e(\theta) = \begin{bmatrix} R(\theta_2) & \begin{bmatrix}\theta_1 + \theta_3\cos(\theta_2) \\ \theta_3\sin(\theta_2)\end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

(b) To compute body manipulator Jacobian, we will use the formula:

$$J_b = \begin{bmatrix} \mathrm{Ad}^{-1}_{g_2 g_3} J_{b,1} & \mathrm{Ad}^{-1}_{g_3} J_{b,2} & J_{b,3} \end{bmatrix}$$

with

$$J_{b,i} = \left(g_i^{-1} \frac{\partial g_i}{\partial \theta_i} \right)^\vee$$

Computing each body Jacobian gives us:

$$J_{b,1}(\theta_1) = \left(g_1^{-1}\frac{\partial g_1}{\partial \theta_1}\right)^\vee$$
$$= \begin{bmatrix} I & -\begin{bmatrix}\theta_1\\0\end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & \begin{bmatrix}1\\0\end{bmatrix} \\ 0 & 0 \end{bmatrix}^\vee = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^\vee = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$J_{b,2}(\theta_2) = \left(g_2^{-1}\frac{\partial g_2}{\partial \theta_2}\right)^\vee$$
$$= \begin{bmatrix} R^\top(\theta_2) & \begin{bmatrix}0\\0\end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -\sin(\theta_2) & -\cos(\theta_2) & 0 \\ \cos(\theta_2) & -\sin(\theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix}^\vee$$
$$= \begin{bmatrix} \cancel{-\cos(\theta_2)\sin(\theta_2) + \sin(\theta_2)\cos(\theta_2)} & -\cos(\theta_2)^2 - \sin(\theta_2)^2 & 0 \\ \sin(\theta_2)^2 + \cos(\theta_2)^2 & \cancel{\sin(\theta_2)\cos(\theta_2) - \cos(\theta_2)\sin(\theta_2)} & 0 \\ 0 & 0 & 0 \end{bmatrix}^\vee$$
$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}^\vee = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$J_{b,3} = \left(g_3^{-1}\frac{\partial g_3}{\partial \theta_3}\right)^\vee$$
$$= \begin{bmatrix} I & -\begin{bmatrix}\theta_3\\0\end{bmatrix} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}^\vee$$
$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Next, accounting for the adjoints:

$$\begin{aligned}
\mathrm{Ad}_{g_2g_3}^{-1} J_{b,1} &= (g_2g_3)^{-1} J_{b,1} (g_2g_3) \\
&= g_3^{-1} g_2^{-1} J_{b,1} g_2 g_3 \\
&= \underbrace{\underbrace{\begin{bmatrix} I & -\begin{bmatrix}\theta_3 \\ 0\end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R^\top(\theta_2) & \begin{bmatrix}0\\0\end{bmatrix} \\ 0 & 1 \end{bmatrix}}_{\begin{bmatrix} R^\top(\theta_2) & 0 \\ 0 & 1 \end{bmatrix}} \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\begin{bmatrix} 0 & 0 & \cos(\theta_2) \\ 0 & 0 & -\sin(\theta_2) \\ 0 & 0 & 0 \end{bmatrix}} \begin{bmatrix} R(\theta_2) & \begin{bmatrix}0\\0\end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} I & \begin{bmatrix}\theta_3 \\ 0\end{bmatrix} \\ 0 & 1 \end{bmatrix}}_{\begin{bmatrix} 0 & 0 & \cos(\theta_2) \\ 0 & 0 & -\sin(\theta_1) \\ 0 & 0 & 0 \end{bmatrix}}^\vee \\
&= \begin{bmatrix} \cos(\theta_2) \\ -\sin(\theta_2) \\ 0 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\mathrm{Ad}_{g_3}^{-1} J_{b,2} &= g_3^{-1} J_{b,2} g_3 \\
&= \begin{bmatrix} I & -\begin{bmatrix}\theta_3 \\ 0\end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I & \begin{bmatrix}\theta_3 \\ 0\end{bmatrix} \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & \theta_3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & \theta_3 \\ 0 & 0 & 0 \end{bmatrix}^\vee = \begin{bmatrix} 0 \\ \theta_3 \\ 1 \end{bmatrix}
\end{aligned}$$

As mentioned last week, a shortcut for this adjoint computation is $\mathrm{Ad}_g = \begin{bmatrix} R & [d]_\times R \\ 0 & R \end{bmatrix}$

Putting everything together gives us:

$$J(\theta) = \begin{bmatrix} \cos(\theta_2) & 0 & 1 \\ -\sin(\theta_2) & \theta_3 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Note that an equivalent way to obtain this jacobian would have been:

$$J_b = \left[\mathrm{Ad}^{-1}_{e^{\xi_1\theta_1}\ldots e^{\xi_3\theta_3}g_0}\xi_1 \quad \mathrm{Ad}^{-1}_{e^{\xi_2\theta_2}\ldots e^{\xi_3\theta_3}g_0}\xi_2 \quad \mathrm{Ad}^{-1}_{e^{\xi_3\theta_3}g_0}\xi_3\right]$$

with

$$\xi_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \quad \xi_2 = \begin{bmatrix}0\\0\\1\end{bmatrix}, \quad \xi_3 = \begin{bmatrix}1\\0\\0\end{bmatrix}$$

Example MATLAB Code

```
% Helper functions
twist = @(w, p) [cross(-w,p); w];
skew = @(w) [0 -w(3) w(2); w(3) 0 -w(1); -w(2) w(1) 0];
unskew = @(skw) [skw(3,2);skw(1,3);skw(2,1)];
hat = @(twist) [skew(twist(4:6)), twist(1:3); 0 0 0 0];
vee = @(htwist) [htwist(1:3,4); unskew(htwist(1:3,1:3))];
Rz = @(theta) [cos(theta) -sin(theta) 0; ...
               sin(theta) cos(theta) 0; 0 0 1];
Rx = @(theta) [1 0 0; 0 cos(theta) -sin(theta); ...
               0 sin(theta) cos(theta)];

% Symbolic variables
syms t1 t2 t3

% Specify exponential mapping twists for zero configuration
twist1 = [1; 0; 0; 0; 0; 0];
twist2 = [0; 0; 0; 0; 0; 1];
twist3 = [1; 0; 0; 0; 0; 0];
g0 = [eye(3), [0;0;0]; 0 0 0 1];

%Compute body twists using method of adjoints
twistb1 = vee(inv(expm(hat(twist1)*t1)* ...
            expm(hat(twist2)*t2)* ...
            expm(hat(twist3)*t3)*g0)* ...
            hat(twist1)* ...
            expm(hat(twist1)*t1)*expm(hat(twist2)*t2)* ...
            expm(hat(twist3)*t3)*g0)
twistb2 = vee(inv(expm(hat(twist2)*t2)* ...
            expm(hat(twist3)*t3)*g0)* ...
            hat(twist2)* ...
            expm(hat(twist2)*t2)*expm(hat(twist3)*t3)*g0)
twistb3 = vee(inv(expm(hat(twist3)*t3)*g0)* ...
            hat(twist3)*expm(hat(twist3)*t3)*g0)

% Simplify body manipulator jacobian
J = simplify([twistb1, twistb2, twistb3])
```

(c) We are going to evaluate the body manipulator Jacobian for two cases:

Case 1: $\theta = (1, \pi/2, 1)^\top$

$$J_b(\theta) = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Here $\det(J_b(\theta)) = -1 \neq 0$, so the matrix has full rank (number of linearly independent columns = 3). Thus, we have full (local) control and this configuration is *not* singular.

Case 2: $\theta = (1, 0, 1)^\top$

$$J_b(\theta) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

The second and third rows are identical, so $\det(J_b(\theta)) = 0$ and the matrix is *not* full rank (rank 2). We have lost control of the system locally, so this configuration *is* singular.

(d) General case: $\theta = (1, \theta_2, 1)^\top$

$$J_b(\theta) = \begin{bmatrix} \cos(\theta_2) & 0 & 1 \\ -\sin(\theta_2) & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

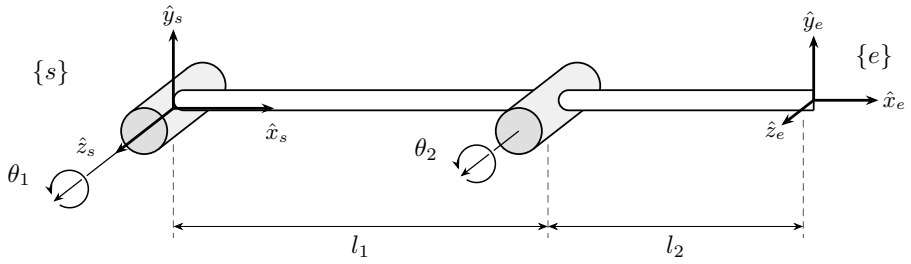
Expanding along the third column, the determinant is

$$\det(J_b(\theta)) = 1 \cdot \det \begin{bmatrix} -\sin(\theta_2) & 1 \\ 0 & 1 \end{bmatrix} = -\sin(\theta_2)$$

The matrix loses rank whenever $\sin(\theta_2) = 0$, that is, for $\theta_2 = k\pi$ with $k \in \mathbb{Z}$ ($\theta_2 = \ldots, -\pi, 0, \pi, 2\pi, \ldots$). Physically, these are the configurations where link 2 lies along the $\hat{x}_s$ axis, so both prismatic joints slide in the same direction. The determinant approaches 0 as we approach these singularities.

Problem 2: Planar RR Manipulator

Consider the planar manipulator below, shown in its zero configuration, with link lengths l_1 and l_2.



(a) Use the product of Lie groups to find the forward kinematics $g_e(\theta)$.

(b) This arm has only two joints, so consider only the position $p_e(\theta)$ of the end-effector. Compute the Jacobian of $p_e(\theta)$.

(c) Evaluate this Jacobian at $\theta = (\theta_1, 0)^\top$ and at $\theta = (\theta_1, \pi/2)^\top$. Is either configuration singular?

Solution

(a) For this manipulator, our homogeneous transformation matrices are:

$$g_1(\theta_1) = \begin{bmatrix} R(\theta_1) & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}, \quad g_2(\theta_2) = \begin{bmatrix} R(\theta_2) & \begin{bmatrix} l_1 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}, \quad g_3(\theta_3) = \begin{bmatrix} I & \begin{bmatrix} l_2 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

Putting these together, the forward kinematics is given by:

$$\begin{aligned} g_e(\theta) &= g_1(\theta_1) g_2(\theta_2) g_3(\theta_3) \\ &= \begin{bmatrix} R(\theta_1 + \theta_2) & \begin{bmatrix} l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) \\ l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) \end{bmatrix} \\ 0 & 1 \end{bmatrix} \end{aligned}$$

(b) This is a kinematically insufficient arm ($m < n$, $2 < 3$). This means that we can't expect total control. One solution is to only consider the outputs being x and y coordinates. This reduces the forward kinematics to:

$$p_e(\theta) = \begin{bmatrix} l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) \\ l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) \end{bmatrix}$$

Since we deal with $p_e(\theta)$ only and not $g_e(\theta)$, there is no orientation to worry about, therefore, the standard Jacobian will be sufficient for our singularity analysis:

$$J_{pos}(\theta) = \frac{\partial p_e(\theta)}{\partial \theta} = \begin{bmatrix} -l_1 \sin(\theta_1) - l_2 \sin(\theta_1 + \theta_2) & -l_2 \sin(\theta_1 + \theta_2) \\ l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) & l_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

(c) Lets consider the following two cases:

Case 1: $\theta = (\theta_1, 0)^\top$

$$J(\theta) = \begin{bmatrix} -(l_1 + l_2)\sin(\theta_1) & -l_2 \sin(\theta_1) \\ (l_1 + l_2)\cos(\theta_1) & -l_2 \cos(\theta_1) \end{bmatrix}$$

This gives two parallel translation vectors. $J(\theta_1, 0)$ does not have full rank (this is a singularity).

Case 2: $\theta = (\theta_1, \pi/2)^\top$

$$J(\theta) = \begin{bmatrix} -l_1 \sin(\theta_1) - l_2 \cos(\theta_1) & -l_2 \sin(\theta_1) \\ l_1 \cos(\theta_1) - l_2 \sin(\theta_1) & l_2 \cos(\theta_1) \end{bmatrix}$$

This gives two non-parallel translation vectors. so $J(\theta_1, \pi/2)$ has full rank (no singularity).
Note: the simplification of $J(\theta)$ above comes from the identities:

$$\sin(\theta_1 + \theta_2) = \sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2)$$
$$\cos(\theta_1 + \theta_2) = \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)$$